

GULBENE, Latvia

WMO: 263480

Lat: 57.1322N

Lon: 26.7189E

Elev: 468

StdP: 14.45

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-0.4	-10.9	3.1	-5.3	-5.1	4.3	0.5	18.5	28.9	16.6	27.4	3.8	310	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	81.8	67.4	78.5	65.2	75.5	63.7	69.5	78.7	67.3	75.3	65.4	72.8	5.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	98.0	74.2	64.4	92.0	71.3	62.5	85.9	69.3	33.8	78.9	32.0	75.4	30.5	72.7	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	14.0	12.2	DB	-11.8	86.4	6.5	2.9	-16.5	88.5	-20.3	90.2	-24.0	91.9	-28.7	94.0	(4)
(5)			WB	-12.1	72.3	6.3	2.2	-16.6	73.9	-20.3	75.2	-23.9	76.5	-28.5	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	21.9	22.7	30.9	43.4	53.1	59.6	64.1	61.5	52.4	42.1	32.9	25.7		
	DBStd	16.93	10.67	11.63	7.38	8.20	7.76	6.38	6.00	5.30	5.82	6.96	8.29	10.47		
	HDD50	4084	871	763	591	232	56	5	0	42	258	513	753			
	HDD65	8284	1336	1183	1056	649	376	184	91	138	379	711	963	1218		
	CDD50	1397	0	0	0	34	152	293	437	355	114	12	0	0		
	CDD65	122	0	0	0	0	7	23	63	28	1	0	0	0		
	CDH74	959	0	0	0	2	64	180	469	236	8	0	0	0		
	CDH80	169	0	0	0	0	6	20	106	37	0	0	0	0		
(14) Wind	WSAvg	5.9	6.7	6.7	6.9	6.4	5.6	5.0	4.7	4.6	5.2	6.0	6.7	6.6		
(15) Precipitation	PrecAvg	26.2	1.9	1.6	1.5	1.3	2.3	3.0	3.0	2.7	2.2	2.8	2.1	1.8		
	PrecMax	31.6	3.3	2.7	3.7	3.7	4.3	5.7	8.7	5.3	3.7	4.6	3.1	3.1		
	PrecMin	20.0	0.7	0.6	0.6	0.2	0.7	0.8	0.4	0.0	0.7	1.3	0.1	1.3		
	PrecStd	3.1	0.7	0.6	0.8	0.8	1.1	1.2	1.9	1.5	0.9	0.9	0.8	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	42.7	53.6	72.7	80.4	82.7	87.1	84.2	75.3	61.1	50.5	46.3		
		MCWB	38.9	38.4	42.7	55.8	63.7	67.1	71.1	69.0	61.7	54.4	48.0	44.4		
	2%	DB	37.1	39.7	48.0	67.4	75.0	79.0	83.2	79.7	69.0	57.0	47.6	42.9		
		MCWB	35.8	37.1	39.9	52.0	60.9	65.0	68.8	65.6	59.4	53.2	45.8	41.5		
	5%	DB	35.5	37.6	43.7	62.4	70.9	75.5	79.2	76.3	65.3	54.4	45.6	40.3		
		MCWB	34.3	35.7	38.3	49.9	57.9	63.7	67.0	63.8	57.7	51.5	44.1	39.1		
	10%	DB	34.1	36.0	40.5	57.7	67.3	71.8	75.8	72.8	62.0	52.0	43.3	38.0		
		MCWB	33.2	34.5	36.6	47.6	56.1	61.7	65.1	62.4	55.9	49.5	42.0	36.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	40.0	44.3	57.2	66.1	70.0	72.7	72.2	64.0	56.0	48.9	45.2		
		MCDB	39.8	41.7	51.0	69.5	76.9	78.8	83.6	81.6	69.1	59.2	50.1	46.4		
	2%	WB	36.1	37.4	41.4	53.8	62.7	67.2	70.2	68.2	60.9	53.8	46.2	41.5		
		MCDB	36.7	38.8	46.0	63.5	71.7	75.7	80.2	75.9	66.1	56.1	47.5	42.5		
	5%	WB	34.6	36.1	39.0	51.2	60.1	64.9	68.2	65.6	58.9	51.9	44.2	39.3		
		MCDB	35.3	37.4	42.7	60.4	67.9	73.3	76.7	72.9	64.4	54.3	45.4	40.2		
	10%	WB	33.3	34.6	37.1	48.8	57.7	62.9	66.3	63.6	56.8	49.9	42.1	36.7		
		MCDB	34.0	35.8	40.1	56.7	65.2	69.8	73.6	70.5	60.9	51.8	43.2	37.7		
(35) Mean Daily Temperature Range	MDBR		7.6	9.6	11.5	16.1	18.1	16.4	16.1	16.9	13.7	9.9	6.8	6.8		
	5% DB	MCDBR	6.3	8.3	15.9	23.6	23.2	21.5	21.0	22.5	18.9	12.5	7.8	6.4		
		MCWBR	6.1	7.0	10.4	11.8	11.5	10.1	9.1	9.9	10.4	8.9	7.2	6.3		
	5% WB	MCDBR	6.1	7.8	13.6	20.3	19.9	18.7	18.9	19.3	16.1	11.6	7.7	6.4		
		MCWBR	6.1	7.0	9.9	10.6	10.9	10.1	9.0	9.6	10.1	9.1	7.3	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.309	0.339	0.380	0.373	0.369	0.398	0.396	0.362	0.338	0.318	0.273		
	taud		2.147	2.141	2.187	2.273	2.364	2.402	2.337	2.339	2.411	2.404	2.246	2.152		
	Ebn at Noon		185	229	255	264	274	277	264	256	248	222	168	151		
	Edh at Noon		17	27	33	36	35	34	36	34	27	21	16	13		
(44) All-Sky Solar Radiation	RadAvg		150	383	825	1252	1671	1736	1650	1326	858	391	133	89		
	RadStd		33	71	111	155	171	140	170	128	94	63	21	17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page